

WEEK 3 PDF REPORT

Week 3 Report - Advanced Security Implementation

User Management System Security Enhancement

Name: Talha

Role: Cybersecurity Intern

Project: Secure User Management System

1. OBJECTIVE

The objective of Week 3 was to perform advanced security testing, implement logging mechanisms, and verify the overall security of the User Management System.

2. TASKS COMPLETED

2.1 Security Testing

- Tested JWT protected routes using Postman
- Verified unauthorized access is blocked
- Valid token allows access to user data

2.2 Logging System

- Implemented Winston logging library
- Logs stored in security.log file
- Application events tracked successfully

2.3 Security Improvements

- JWT authentication verified
- Password hashing using bcrypt

- Protected routes secured using middleware
- Input validation maintained



3. TESTING RESULTS

- Valid token → Access granted
- Invalid token → Access denied (401)
- No token → Access denied (401)
- System behaves securely in all cases



4. FINAL CHECKLIST

- Input validation implemented
- Passwords hashed securely
- JWT authentication working
- Logging system active
- Protected routes verified



5. CONCLUSION

Week 3 completed advanced security testing and final validation of the system. The application is now secure, tested, and production-ready at a basic level.